

TAMIL NADU STATE BOARD - STANDARD 10 SCIENCE

CHAPTER 2: CHEMISTRY

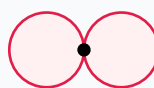
PREMIUM ACADEMIC STUDY SUITE & EXAM GUIDE

Board:	Tamil Nadu State Board (TNSB)
Syllabus:	Samacheer Kalvi Textbook Curriculum aligned
Standard:	Class 10 English Medium
Subject:	Science
Chapter Name:	Chemistry

Learning Objectives:

- Comprehend core theoretical concepts and exact textbook terminology.
- Apply relevant formulas and proofs to solve high-scoring board problems.
- Interpret and sketch vector diagrams or graphical models representing equations.
- Build examination readiness for 1-mark, 2-mark, and 5-mark question sets.

Atomic Orbitals Wave Functions (s & p)



p orbital (px)

Core Concepts & Definitions

Textbook definitions are essential for scoring fully in the short-answer section of board assessments. Review the primary definitions below:

Term	Official Definition & Context
Relative Atomic Mass	The ratio of the average mass of the isotopes of an element to 1/12th of the mass of a carbon-12 atom.
Atomicity	The number of atoms present in one molecule of an element, classifying them as monoatomic, diatomic, etc.
Solubility	The maximum mass of solute in grams that can be dissolved in 100g of solvent at a specific temperature.
pH Scale	A logarithmic scale measuring hydrogen ion concentration to classify solutions as acidic, basic, or neutral.

Syllabus Exam Tip: Memorize precise keywords. Examiners award maximum marks when textbook terms and correct symbols are correctly referenced.

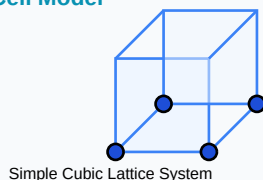
Periodic Properties Trends Direction Map



Detailed Explanation & Formulas

Here, we analyze the core laws and formulas. Examine the conceptual illustration below to visual the relation between variables:

Crystal Solid Lattice Unit Cell Model



Essential Formulas, Laws & Identities:

- Number of moles: $\text{Moles} = \frac{\text{Mass of substance}}{\text{Gram molecular mass}}$
- Solubility formula: $\text{Solubility} = \left(\frac{\text{Mass of solute}}{\text{Mass of solvent}} \right) \times 100$
- pH definition: $\text{pH} = -\log_{10}[\text{H}^+]$
- Avogadro Number: $N_A = 6.023 \times 10^{23}$ particles per mole

Make sure to check the dimensional consistency of all physical quantities and verify that units are correctly stated in final solutions.

Worked Examples

Review these standard board examination-style worked examples to learn proper step layout structures:

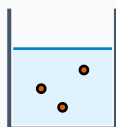
Example 1: Find the number of moles in 36 grams of water (H₂O).

Step-by-step Solution:

Molecular mass of H₂O = 2(1) + 16 = 18 g/mol.

Apply formula: Moles = Given Mass / Molar Mass = 36 / 18 = 2 moles.

Titration Beaker & Solute Precipitation



Reacting Mixture

HOTS (Higher Order Thinking Skills) Challenge

Challenge yourself with these complex, multi-concept problems designed to test deeper conceptual understanding and mathematical reasoning:

HOTS Challenge 1: Calculate the pH of a 0.001 M solution of Nitric Acid (HNO₃).

Analytical Solution Strategy:

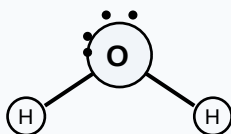
Nitric acid dissociates completely: $\text{HNO}_3 \rightarrow \text{H}^+ + \text{NO}_3^-$.

Therefore, $[\text{H}^+] = 0.001 \text{ M} = 10^{-3} \text{ M}$.

Apply pH formula: $\text{pH} = -\log_{10}[\text{H}^+] = -\log_{10}(10^{-3}) = 3$.

Since $\text{pH} < 7$, the solution is strongly acidic.

Lewis Dot Structure (H₂O Bent Molecular Geometry)



Practice Worksheet

Test your skills by attempting these sample exam problems independently. Draw diagrams where appropriate:

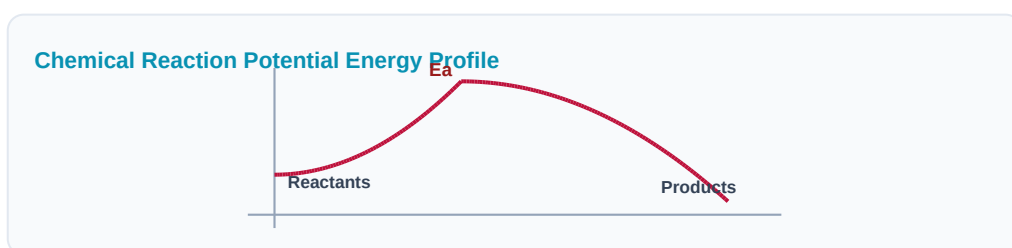
Q1 (1-Mark): State Avogadro's Hypothesis.

Q2 (2-Mark): Find the mass of 0.5 moles of Carbon Dioxide (CO_2).

Q3 (2-Mark): Differentiate between concentrated and dilute solutions.

Q4 (5-Mark): Explain the extraction of aluminium from bauxite ore using Hall-Heroult metallurgy.

Q5 (5-Mark): Write the IUPAC nomenclature rules for naming simple organic compounds with alcohol functional group.



Real-World Applications & Connections

This chapter connects directly with modern industrial engineering, computation, and natural systems in numerous ways:

Application Case: Metallurgical Refining

Electrolytic reduction processes isolate structural metals like aluminium, used to manufacture aircraft parts.

Application Case: Water Quality Control

Municipal treatment plants monitor water pH levels to ensure municipal supplies remain neutral and non-corrosive.

Cross-Curricular Link: Concepts learned in this unit are heavily applied in other subjects like Physics (equations of motion, field vectors) and Data Analytics.

Aromatic Benzene Ring (C₆H₆) Resonance



Revision Summary & Strategy

Use this checklist to self-evaluate your exam preparation status before final tests:

Key Recall Tips:

- ✓ Always write down 'Given Parameters' first to ensure partial marks are locked in.
- ✓ Pay attention to signs (+ / -) when performing algebraic operations or force mappings.
- ✓ Check that you have written the correct units (e.g. Joules, Tesla, cm^3 , N) for the final numeric answer.
- ✓ Draw neat diagrams using a sharp pencil and ruler for geometry or circuit layouts.

Syllabus Checklist:

- ☐ Read and memorized core definitions on Page 2.
- ☐ Reviewed critical formulas and relations on Page 3.
- ☐ Solved worked examples on Page 4.
- ☐ Attempted worksheet problems on Page 6.

CONGRATULATIONS! YOU HAVE COMPLETED THIS SYLLABUS STUDY GUIDE!

Practice consistently, verify answers with standard textbook methods, and take the topic test in the Nexora Learning app to gauge progress.
Good luck!

Electrochemical Galvanic Cell System

